



EEMG Department
SIMHADRI

Note No :1



Ref:SMTTP/EEMG/2025-26/LMI 252/192109

Date:03/06/2025(10:29:28)

Subject:Revision of LMI for Determination of specific water consumption

LMI Reference No:LMI/OIN/OPS/SYST/034

LMI is revised on the basis of COS-ISO-00-OIN/OPS/SYST/034 Rev :03 April 2025.

Submitted for apporval

Submitted by:

नीरज कुमार सिंह - वरिष्ठ प्रबंधक

NEERAJ KUMAR SINGH - 103459 (SENIOR MANAGER ,EEMG) ,Simhadri
(NKSINGH10@NTPC.CO.IN, 9471007277)

On:03/06/2025(10:29:28) **Ref:**SMTTP/EEMG/2025-26/LMI 252/192109

Note No:2

LMI has been revised as per the revised OIN.

LMI is in line with the OIN. May please be approved.

Submitted by:

नवीन किशोर - अपर महाप्रबंधक

Navin Kishore - 006964 (ADDL. GENERAL MANAGER ,EEMG)

(NAVINKISHORE@NTPC.CO.IN, 9650990460) ,Simhadri

On:03/06/2025(10:36:28) **Ref:**SMTTP/EEMG/2025-26/LMI 252/192109

Note No:3

Checked and found OK.

Submitted by:

सोमनाथ भट्टाचार्य - अपर महाप्रबंधक

Somnath Bhattacharya - 007203 (ADDL. GENERAL MANAGER ,OPERATION)

(SOMNATHBHATTACHARYA@NTPC.CO.IN, 9431600725) ,Simhadri

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Note No:4

Checked and found ok.

Submitted for approval please.

Submitted by:

सांतनु जेना - उप महाप्रबंधक (रसायन)

Santanu Jana - 007743 (DEPUTY GENERAL MANAGER (CHEMIST) ,CHEMISTRY)
(SANTANUJANA@NTPC.CO.IN, 7458012612) ,Simhadri

On:06/06/2025(15:20:59) **Ref:**SMTPP/EEMG/2025-26/LMI 252/192109

Note No:5

Submitted by:

अजय गिरि - अपर महाप्रबंधक

Ajay Giri - 007629 (ADDL. GENERAL MANAGER ,FGD)
(AJAYGIRI@NTPC.CO.IN, 7440228332) ,Simhadri

On:06/06/2025(15:35:57) **Ref:**SMTPP/EEMG/2025-26/LMI 252/192109

Note No:6

Checked and found OK.

Submitted by:

अनिल कुमार मौर्या - अपर महाप्रबंधक

Anil Kumar Maurya - 008631 (ADDL. GENERAL MANAGER ,C & I MAINT)
(ANILKUMARMAURYA@NTPC.CO.IN, 8004940574) ,Simhadri

On:17/06/2025(21:19:16) **Ref:**SMTPP/EEMG/2025-26/LMI 252/192109

Note No:7

Submitted by:

प्रसन्ता कुमार जेना - महाप्रबंधक

Prasanta Kumar Jena - 006081 (GENERAL MANAGER ,O & M)
(PKJENA@NTPC.CO.IN, 9650990836) ,Simhadri

On:20/06/2025(19:42:57) **Ref:**SMTPP/EEMG/2025-26/LMI 252/192109

Note No:8

LMI has been revised as per the revised OIN.

LMI is in line with the OIN. May please be approved.

Submitted by:

नवीन किशोर - अपर महाप्रबंधक

Navin Kishore - 006964 (ADDL. GENERAL MANAGER ,EEMG)
(NAVINKISHORE@NTPC.CO.IN, 9650990460) ,Simhadri

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Note No:9

Submitted by:

प्रसन्ता कुमार जेना - महाप्रबंधक

Prasanta Kumar Jena - 006081 (GENERAL MANAGER ,O & M)
(PKJENA@NTPC.CO.IN, 9650990836) ,Simhadri

On:25/06/2025(17:19:08) **Ref:**SMTTPP/EEMG/2025-26/LMI 252/192109

Note No:10

Approved & Submitted by:

समीर शर्मा - कार्यकारी निदेशक

Sameer Sharma - 005041 (EXECUTIVE DIRECTOR ,HEAD OF PROJECT)
(SAMEERSHARMA@NTPC.CO.IN, 9650990022) ,Simhadri

On:25/06/2025(18:45:52) **Ref:**SMTTPP/EEMG/2025-26/LMI 252/192109

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NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

Determination of specific water consumption

LMI/OIN/OPS/SYST/034 Rev:04

MAY-2025

Approved for Implementation by: **HOP (SIMHADRI)**

LMI APPROVAL DOCUMENT

1	LMI NAME	Determination of specific water consumption
2	LMI No.	LMI/OIN/OPS/SYST/034 Rev:04
3	Reference Document	COS-ISO-00 -OIN/OPS/SYST/034/ Rev:03 April:2025
4	SupersededDocument	LMI/OIN/OPS/034 Rev:03 April 22
5	Prepared By	Name: Neeraj Kumar Singh Sr. Manager (EEMG)
6	Checked By	Name: Navin Kishore AGM (EEMG)
7	Reviewed By	Name: P. K Jena G M (O&M)
8	Approved By	Name: Sameer Sharma H O P (Simhadri)

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TITLE	Determination of specific water consumption
LMI No.	LMI/OIN/OPS/SYST/034 Rev:04

1.0 INTRODUCTION

Large quantity of water is used in thermal power stations for cooling in condensers of steam turbine. It is either used in once through mode or in close cycle mode. In once through mode, whatever water is taken from water body is given back to water body, excepting little quantity consumed in evaporation. In close cycle mode, heat picked up in condenser is evacuated to atmosphere through cooling towers. In such cases, continuous addition of make-up water is required to take care of the evaporation as well as drift losses occurring in cooling towers. Also, a part of re-circulating water is replenished with makeup water to maintain the salt level to acceptable limits.

In addition to consumption in CW/CT circuit, large quantity of water is consumed for wet disposal of ash, in production of de-mineralized water, as makeup in the cooling circuit of various coolers like lub oil coolers, generator coolers, compressors etc., in housekeeping and firefighting applications. Water is also consumed for offices and for domestic application in township etc.

Water being a scarce resource all efforts are to be made to minimize the consumption of water. This objective is achieved through proper selection and design of power plant schemes / systems at initial stage itself and by following proper operation and maintenance practices. Audit of water balance system also helps in identifying the wastage / inefficient utilization of water. Such audits compare the present flows in different water circuits and compare it with the designed flows, which help in identifying the problematic areas / systems. All stations to develop action plan based on recommendations of the water balance audit and try to bring their water consumption within the norms specified as per Environment (Protection) Amendment Rules 2015.

As a first step towards systematic efforts for minimization of water consumption, it is essential to periodically measure / asses total water consumption and specific water consumption of the station. Regular monitoring and trending of these parameters will help in quick identification of any deterioration. It will also help in assessment of impact of any modification / measures taken for water conservation.

TITLE	Determination of specific water consumption
LMI No.	LMI/OIN/OPS/SYST/034 Rev:04

Simhadri two types of water being used:

Sea water: for condenser cooling and ash disposal. This sea water is drawn from Bay of Bengal through 3 no of Sea water pumps. The hot water from condenser is cooled in NDCTs and part of it is used for ash disposal and the remaining water is sent back to sea through EDP (Effluent Disposal Pumps)

Sweet Water: for process requirement and Potable purpose. Sweet water is drawn from Yeleru canal and is pumped to reservoir through sweet water pumps. From reservoir required water is drawn and is processed in PT plant. This processed water is supplied as DM water, Clarifier water and Potablewater.

2.0 SUPERSEDED DOCUMENTS

LMI/OIN/OPS/SYST/034 Rev: 03, Apr:2022

3.0 DETAILS OF REVISION

Entire LMI has been revised according to TC document revision.

Following new points are added

1. Clause No. 4.4 and 4.5 added
2. Clause No. 5.7, 5.8 and 5.9 added.

4.0 OBJECTIVES

4.1 For Simhadri consent quantities are:

Sweet water21.6 MLD (Million Litres per Day)
Sea water.....429.24 MLD

4.2 This LMI will help to monitor the Specific water consumption and maintain water consumption within consent limits.

4.3 Determination of specific water consumption on withdrawal (from natural resource) basis of the station is to be done on monthly, quarterly, and yearly (cumulative) basis.

Resp.: HOD (EEMG)

4.4 Determination of Specific water on industrial consumption basis of the station is to be done on monthly, quarterly and yearly (cumulative) basis.

Resp.: HOD (EEMG)

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- 4.5** Monitoring and metering of domestic water consumption in township is to be done on monthly, quarterly and yearly (cumulative) basis.

Resp.: HOD (EEMG)

5.0 METHODOLOGY

- 5.1** Flow meters have been installed at all consumptions points as indicated in Annexures-I & II. All points of water consumption are listed, and water flows are being recorded against each withdrawal point and summed up to get total water consumption of the station (in M³/hr).

- 5.2** The flow meter readings have to be recorded on daily basis and consumption on daily/monthly/yearly basis shall be derived.

Resp.: HOD (EEMG/OPN)

- 5.3** Determination of specific water consumption (on withdrawal basis): Recording of withdrawal water flows at reservoir inlet as well as consumptive water flows at Raw Water Pump House outlet are to be recorded on daily /monthly basis. Specific water consumption is the ratio of total water withdrawal from natural source (in M3) and total generation (in MWhr).

Resp.: HOD (EEMG)

- 5.4** Determination of specific water Industrial consumption: The total consumptive water includes the water withdrawn from the reservoir and storage losses due to evaporation and seepage in reservoir excluding township water consumption. Specific water industrial consumption is the ratio of total industrial water (M3) and total generation (MWhr).

Resp.: HOD(EEMG)

- 5.5** Stations to install water meters for measurement of water drawl from all sources.

These meters should be connected to station PI system / DDCMS for display, trending, and archiving.

Resp.: HOD (C&I)

- 5.6** The township water from plant to township is to be metered and monitored. The CSR water going from plant is to be metered and recorded separately, and it should not be included in industrial water consumption.

Resp.: HOD (C&I/ CHEMISTRY)

- 5.7** FGD water consumption to be measured separately and recorded.

Resp.: HOD(OPN/EEMG)

- 5.8** All water flow meters used for specific water consumption need to be cross validated with ultrasonic portable water flow meter on annual basis Specific Water Consumption shall be calculated on daily / monthly basis.

Resp.: HOD(EEMG)

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5.9 Station logs the water data in SAP by 3rd of following month as per CenPEEP IOM dated 29.06.2023.
Resp.: HOD(EEMG)

6.0 REVIEW

This LMI shall be reviewed by HOP (Simhadri) every 2 years or as and when necessary.

Enclosure:

Check list

Annexure-I

Annexure-II

TITLE	Determination of specific water consumption
LMI No.	LMI/OIN/OPS/SYST/034 Rev:04

Specific Water Consumption Report Format

Sweet Water

Sr No	Month	Monthly Actual Gross Generation (MWh)	Monthly Raw Water Withdrawal from Natural Source (M3)	Monthly Actual Industrial Water Consumption (M3)	Monthly Actual drinking Water Consumption (M3) (Township)	SWC Withdrawal (M3/MWh)	SWC (Consumption-Industrial) (M3/MWh)
1	APR						
2	MAY						
3	JUN						
4	JUL						
5	AUG						
6	SEP						
7	OCT						
8	NOV						
9	DEC						
10	JAN						
11	FEB						
12	MAR						
Total	FY						

TITLE	Determination of specific water consumption
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Specific Water Consumption Report Format

Sea Water

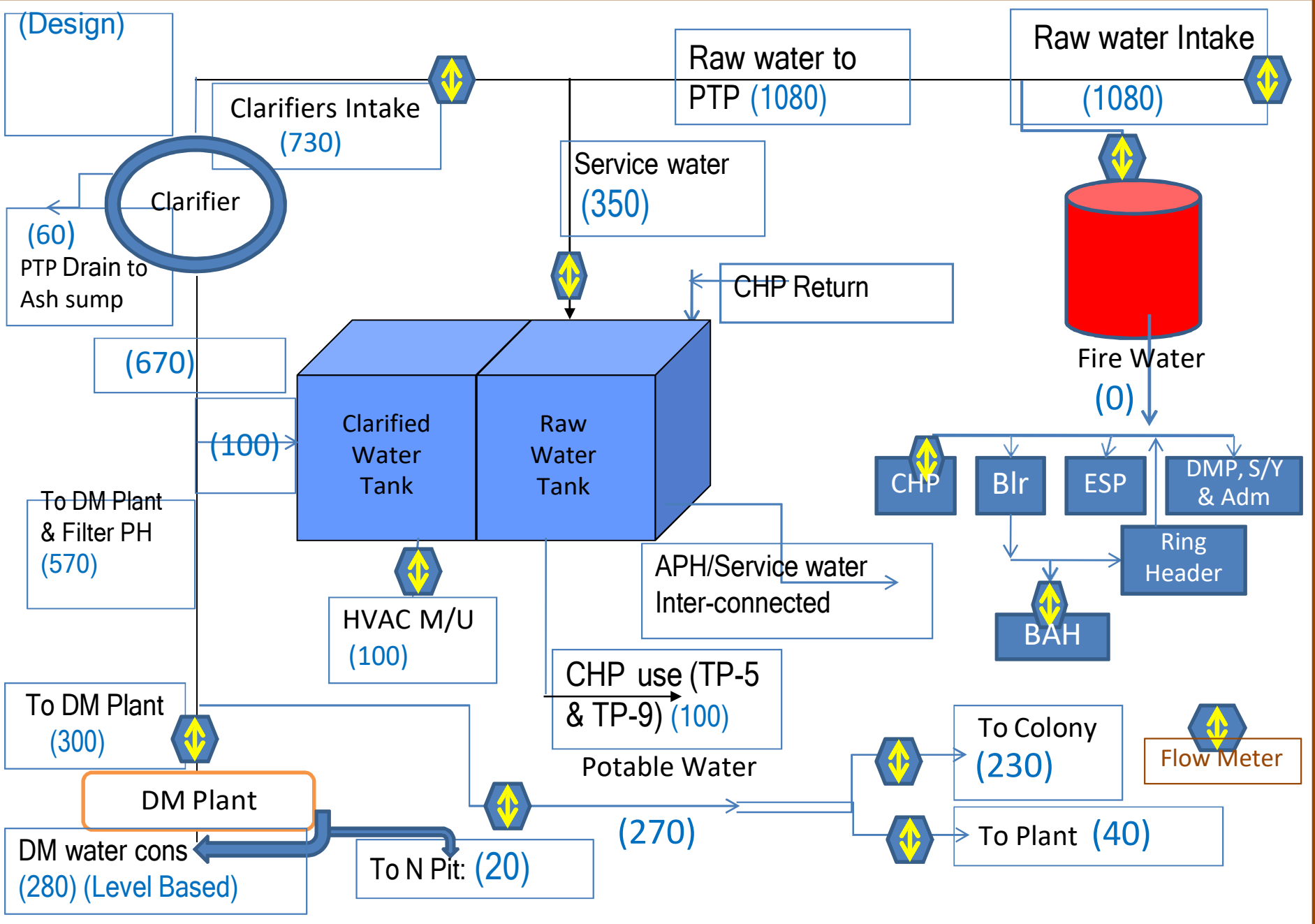
Sr No	Month	Monthly Actual Gross Generation (MWh)	Monthly Sea Water withdrawal (M3)	Sea water used in Ash water Make Up (M3)	Effluent Discharge (M3)	Evaporation Loss (M3)	Sp. water Consumption (M3/Mwh)
1	APR						
2	MAY						
3	JUN						
4	JUL						
5	AUG						
6	SEP						
7	OCT						
8	NOV						
9	DEC						
10	JAN						
11	FEB						
12	MAR						
Total	FY						

Annexure-I

SWEET WATER BALANCE

Values $m^3/hr.$

(Design)



Annexure-II

SEA WATER BALANCE